

Set Theory and Set Algebra

A Complete Chapter for First-Year College Students

1 Learning Goals and Study Plan

Note

After working through this chapter, you should be able to:

- Use precise notation: $\subseteq, \subsetneq, \cup, \cap, \setminus, A^c, \mathcal{P}(A)$.
- Prove subset and set identities using the **element method**.
- Manipulate complements using **De Morgan's laws**.
- Solve computational set algebra problems (including intervals in $\mathbb{R}$).
- Work with power sets and basic counting (cardinality) for finite sets.

Recommended approach:

1. Learn definitions and proof templates (Sections 2–4).
2. Memorize the identities as *tools* by solving the worked problems (Sections 6–9).
3. Attempt practice exercises (Section 10), then check your work.

2 Foundations: Universe, Membership, Equality

Definition 2.1: Universe and complement

A **universe** U is a set containing all objects under discussion. For a set $A \subseteq U$, its **complement** is

$$A^c = U \setminus A = \{x \in U : x \notin A\}.$$

Definition 2.2: Membership and set equality

$x \in A$ means “ x is an element of A ”.

Two sets are equal if and only if they contain exactly the same elements:

$$A = B \iff \forall x (x \in A \iff x \in B).$$

Note

Element method (most important technique). To prove $A \subseteq B$: take an arbitrary x , assume $x \in A$, deduce $x \in B$.

To prove $A = B$: prove both $A \subseteq B$ and $B \subseteq A$.

3 Subset and Proper Subset as Relations

Definition 3.1: Subset and proper subset

For sets A, B :

$$A \subseteq B \iff \forall x (x \in A \Rightarrow x \in B).$$

$$A \subsetneq B \iff (A \subseteq B) \text{ and } (A \neq B).$$

Proposition 3.1: Reflexive, antisymmetric, transitive

For all sets A, B, C :

- (Reflexive) $A \subseteq A$.
- (Antisymmetric) If $A \subseteq B$ and $B \subseteq A$, then $A = B$.
- (Transitive) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Hence $\subseteq$ is a **partial order** on $\mathcal{P}(U)$.

Solution (step-by-step)

Reflexive: Let $x \in A$. Then $x \in A$. So $x \in A \Rightarrow x \in A$. Hence $A \subseteq A$.

Antisymmetric: If $A \subseteq B$ and $B \subseteq A$, then $\forall x, x \in A \Leftrightarrow x \in B$, hence $A = B$.

Transitive: If $x \in A$, then $x \in B$, then $x \in C$, hence $A \subseteq C$.

Proposition 3.2: Properties of proper subset

- (Irreflexive) $A \not\subsetneq A$ for every A .
- (Transitive) If $A \subsetneq B$ and $B \subsetneq C$, then $A \subsetneq C$.

Hence $\subsetneq$ is a **strict partial order** on $\mathcal{P}(U)$.

Solution (step-by-step)

Irreflexive: $A \subsetneq A$ would require $A \neq A$, impossible.

Transitive: From $A \subsetneq B$ and $B \subsetneq C$ we get $A \subseteq B \subseteq C$, hence $A \subseteq C$. Also $B \subsetneq C$ gives an element in C not in B , and since $A \subseteq B$ it is not in A either, so $A \neq C$. Thus $A \subsetneq C$.

4 Operations: Union, Intersection, Difference, Complement

Definition 4.1: Union and intersection

$$A \cup B = \{x \in U : x \in A \text{ or } x \in B\}, \quad A \cap B = \{x \in U : x \in A \text{ and } x \in B\}.$$

Definition 4.2: Difference and symmetric difference

$$A \setminus B = \{x \in U : x \in A \text{ and } x \notin B\}.$$

The **symmetric difference** is

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

Proposition 4.1: Membership dictionary (logic translation)

For any $x \in U$:

$$x \in A \cup B \Leftrightarrow (x \in A) \vee (x \in B), \quad x \in A \cap B \Leftrightarrow (x \in A) \wedge (x \in B),$$

$$x \in A^c \Leftrightarrow x \notin A, \quad x \in A \setminus B \Leftrightarrow (x \in A) \wedge (x \notin B).$$

Solution (step-by-step)

These are direct restatements of the definitions.

5 Set Algebra Laws and De Morgan's Laws

5.1 Core algebra laws

Proposition 5.1: Commutative and associative laws

$$\begin{aligned} A \cup B &= B \cup A, & A \cap B &= B \cap A, \\ (A \cup B) \cup C &= A \cup (B \cup C), & (A \cap B) \cap C &= A \cap (B \cap C). \end{aligned}$$

Solution (step-by-step)

Use the element method. For example:

$$x \in A \cup B \Leftrightarrow (x \in A) \vee (x \in B) \Leftrightarrow (x \in B) \vee (x \in A) \Leftrightarrow x \in B \cup A.$$

The other identities are similar.

Proposition 5.2: Distributive laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Solution (step-by-step)

Let $x \in U$.

$$\begin{aligned} x \in A \cap (B \cup C) &\Leftrightarrow (x \in A) \wedge ((x \in B) \vee (x \in C)) \\ &\Leftrightarrow ((x \in A) \wedge (x \in B)) \vee ((x \in A) \wedge (x \in C)) \Leftrightarrow x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

The second distributive law is proved similarly.

Proposition 5.3: Identity, domination, idempotent, absorption

$$\begin{aligned} A \cup \emptyset &= A, & A \cap U &= A, & A \cup U &= U, & A \cap \emptyset &= \emptyset, \\ A \cup A &= A, & A \cap A &= A, \end{aligned}$$

$$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A.$$

Solution (step-by-step)

Each follows by translating to membership logic. For absorption:

$$x \in A \cup (A \cap B) \Leftrightarrow (x \in A) \vee ((x \in A) \wedge (x \in B)) \Leftrightarrow (x \in A),$$

since $P \vee (P \wedge Q) \Leftrightarrow P$.

5.2 Complement laws and De Morgan

Proposition 5.4: Complement laws

$$A^{cc} = A, \quad A \cup A^c = U, \quad A \cap A^c = \emptyset.$$

Solution (step-by-step)

For $A^{cc} = A$: $x \in A^{cc} \Leftrightarrow x \notin A^c \Leftrightarrow x \in A$.

The other two follow similarly from the meaning of “not in A ”.

Proposition 5.5: De Morgan’s Laws

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

More generally, for a family $\{A_i\}_{i \in I}$:

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c, \quad \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

Solution (step-by-step)

For the two-set union law:

$$x \in (A \cup B)^c \Leftrightarrow x \notin A \cup B \Leftrightarrow \neg((x \in A) \vee (x \in B))$$

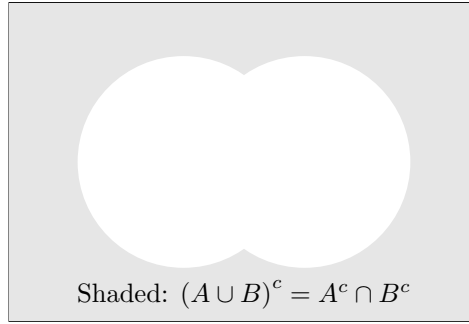
$$\Leftrightarrow (\neg(x \in A)) \wedge (\neg(x \in B)) \Leftrightarrow (x \in A^c) \wedge (x \in B^c) \Leftrightarrow x \in A^c \cap B^c.$$

The intersection law and family versions are analogous.

5.3 Venn diagram (conceptual visualization)

Example 5.1: Venn diagram picture for De Morgan

The shaded region of $(A \cup B)^c$ is everything outside both circles (so it equals $A^c \cap B^c$).



6 Power Set (Concept + Propositions + Counting)

Definition 6.1: Power set

The **power set** of A is the set of all subsets of A :

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

Proposition 6.1: Monotonicity of power set

$$A \subseteq B \iff \mathcal{P}(A) \subseteq \mathcal{P}(B).$$

Solution (step-by-step)

($\Rightarrow$) If $A \subseteq B$ and $S \in \mathcal{P}(A)$, then $S \subseteq A \subseteq B$, hence $S \in \mathcal{P}(B)$. So every element of $\mathcal{P}(A)$ lies in $\mathcal{P}(B)$, i.e. $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

($\Leftarrow$) If $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, then $A \in \mathcal{P}(A)$ (since $A \subseteq A$). Hence $A \in \mathcal{P}(B)$, so $A \subseteq B$.

Proposition 6.2: Cardinality of the power set (finite case)

If A is finite and $|A| = n$, then

$$|\mathcal{P}(A)| = 2^n.$$

Solution (step-by-step)

Each element of A has two choices: either it is included in a subset or not included. With n independent yes/no choices, the number of subsets is 2^n .

7 Cardinality (Finite Sets) and Counting Identities

Definition 7.1: Cardinality

For a finite set A , $|A|$ denotes the number of elements of A .

Proposition 7.1: Inclusion–exclusion (two sets)

For finite sets A, B :

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

If $A \cap B = \emptyset$ (disjoint), then $|A \cup B| = |A| + |B|$.

Solution (step-by-step)

Count $|A| + |B|$: elements in $A \cap B$ are counted twice, so subtract $|A \cap B|$ once to correct the overcount.

Proposition 7.2: Inclusion–exclusion (three sets)

For finite sets A, B, C :

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Solution (step-by-step)

Add the three sizes, subtract pairwise overlaps (they were double-counted), then add back the triple overlap (it was subtracted too many times).

Proposition 7.3: Difference and symmetric difference (finite)

For finite sets:

$$|A \setminus B| = |A| - |A \cap B|, \quad |A \Delta B| = |A| + |B| - 2|A \cap B|.$$

Solution (step-by-step)

A splits disjointly into $(A \setminus B) \cup (A \cap B)$, so $|A| = |A \setminus B| + |A \cap B|$. Rearranging gives the first formula.

Also $A \Delta B = (A \setminus B) \cup (B \setminus A)$ is a disjoint union, so $|A \Delta B| = |A \setminus B| + |B \setminus A|$. Use $|A \setminus B| = |A| - |A \cap B|$ and $|B \setminus A| = |B| - |A \cap B|$.

8 Set Algebra: Worked Problems (Solved)

Problem 1: Simplify using identities

Problem

Simplify the expression:

$$(A \cup B) \cap (A \cup B^c)$$

and express it using only A .

Solution (step-by-step)

Use distributive law: $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$, but here the convenient identity is

$$(A \cup B) \cap (A \cup C) = A \cup (B \cap C).$$

(You can prove it by distributivity.)

Take $C = B^c$:

$$(A \cup B) \cap (A \cup B^c) = A \cup (B \cap B^c) = A \cup \emptyset = A.$$

Problem 2: De Morgan + difference

Problem

Prove the identity:

$$(A \setminus B)^c = A^c \cup B.$$

Solution (step-by-step)

First rewrite the difference:

$$A \setminus B = A \cap B^c.$$

Now take complements and apply De Morgan:

$$(A \setminus B)^c = (A \cap B^c)^c = A^c \cup B^{cc} = A^c \cup B.$$

Problem 3: Symmetric difference identities

Problem

Show that

$$A \Delta B = (A \cup B) \setminus (A \cap B).$$

Solution (step-by-step)

Start from the definition:

$$A \Delta B = (A \setminus B) \cup (B \setminus A) = (A \cap B^c) \cup (B \cap A^c).$$

Now use logic:

$$x \in (A \cap B^c) \cup (B \cap A^c) \iff ((x \in A) \wedge (x \notin B)) \vee ((x \in B) \wedge (x \notin A)).$$

This means: x is in exactly one of A, B , i.e. x is in $A \cup B$ but not in $A \cap B$:

$$\iff ((x \in A) \vee (x \in B)) \wedge \neg((x \in A) \wedge (x \in B)) \iff x \in (A \cup B) \setminus (A \cap B).$$

Thus the sets are equal.

Problem 4: Power set computation

Problem

Let $A = \{1, 2, 3\}$. List $\mathcal{P}(A)$ and verify $|\mathcal{P}(A)| = 2^{|A|}$.

Solution (step-by-step)

All subsets of A are:

$$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$$

So $|\mathcal{P}(A)| = 8$. Since $|A| = 3$, $2^{|A|} = 2^3 = 8$. Verified.

Problem 5: Counting with inclusion–exclusion

Problem

In a class of 60 students, 35 study Physics (set P), 28 study Mathematics (set M), and 15 study both. How many study at least one of Physics or Mathematics? How many study neither?

Solution (step-by-step)

“At least one” means $|P \cup M|$. By inclusion–exclusion:

$$|P \cup M| = |P| + |M| - |P \cap M| = 35 + 28 - 15 = 48.$$

“Neither” means complement in the class universe of 60 students:

$$\text{neither} = 60 - |P \cup M| = 60 - 48 = 12.$$

9 Intervals in $\mathbb{R}$ (Complement, Union, Intersection)

Note

When the universe is $\mathbb{R}$, the complement of an interval becomes a union of rays. Always check endpoints carefully (open vs closed).

Problem

Let $A = [-1, 3)$ and $B = (2, 5]$ in universe $U = \mathbb{R}$. Find $(A \cup B)^c$.

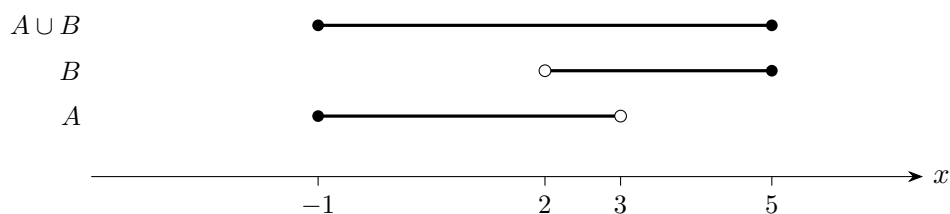
Solution (step-by-step)

First compute $A \cup B$. The sets overlap on $(2, 3)$, and together cover continuously from -1 up to 5 . The endpoint -1 is included (in A). The endpoint 5 is included (in B). Also 3 is included because $3 \in B$. Thus

$$A \cup B = [-1, 5].$$

Therefore

$$(A \cup B)^c = \mathbb{R} \setminus [-1, 5] = (-\infty, -1) \cup (5, \infty).$$



10 Conceptual Problems (with Solutions): Prove / Disprove / Counterexample

Problem 1: “Total comparability” is false

Problem

Disprove the statement: “For any sets A, B , either $A \subseteq B$ or $B \subseteq A$.”

Solution (step-by-step)

Take $A = \{1\}$ and $B = \{2\}$. Then $1 \in A$ but $1 \notin B$, so $A \not\subseteq B$. Also $2 \in B$ but $2 \notin A$, so $B \not\subseteq A$. Thus neither inclusion holds, so the statement is false.

Problem 2: Set equality by two inclusions

Problem

Prove: $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Solution (step-by-step)

$(\Rightarrow)$ If $A = B$, then clearly $A \subseteq B$ and $B \subseteq A$.
 $(\Leftarrow)$ If $A \subseteq B$ and $B \subseteq A$, then by antisymmetry of $\subseteq$, $A = B$.

11 Practice Exercises (No Solutions Here)

Note

Write full solutions using the element method. For simplification problems, show your algebra steps and cite laws (distributive, absorption, De Morgan, etc.).

A. Skill drills (simplify)

1. Simplify: $(A \cap B) \cup (A \cap B^c)$.
2. Simplify: $(A \cup A^c)^c$.
3. Simplify: $(A \cup B) \cap (A^c \cup B)$.
4. Prove: $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
5. Prove: $(A \setminus B) \setminus C = A \setminus (B \cup C)$.

B. De Morgan practice

1. Express $(A \cup B \cup C)^c$ using only complements and intersections.
2. Express $(A \cap B \cap C)^c$ using only complements and unions.
3. Compute $(A^c \cap B^c \cap C^c)^c$ and simplify.

C. Power set and counting

1. List $\mathcal{P}(\{a, b\})$ and compute its cardinality.
2. If $|A| = 5$, compute $|\mathcal{P}(A)|$.
3. In a group of 100 people, 60 like tea (T), 55 like coffee (C), and 30 like both. Find $|T \cup C|$ and the number who like neither.

D. Interval exercises in $\mathbb{R}$

1. Let $A = (0, 2]$ and $B = [4, 6)$. Find $(A \cup B)^c$ in $\mathbb{R}$.
2. Let $A = (-3, 1]$ and $B = [-1, 4)$. Find $(A \cap B)^c$.
3. Let $U = [0, 10]$, $A = [2, 7)$, $B = (5, 9]$. Find $(A \cup B)^c$ relative to U .

Final advice: When stuck, translate membership into logic: “ $x \in \text{LHS}$ ” $\Leftrightarrow$ (logical statement) $\Leftrightarrow$ “ $x \in \text{RHS}$ ”.